

Artificial Intelligence

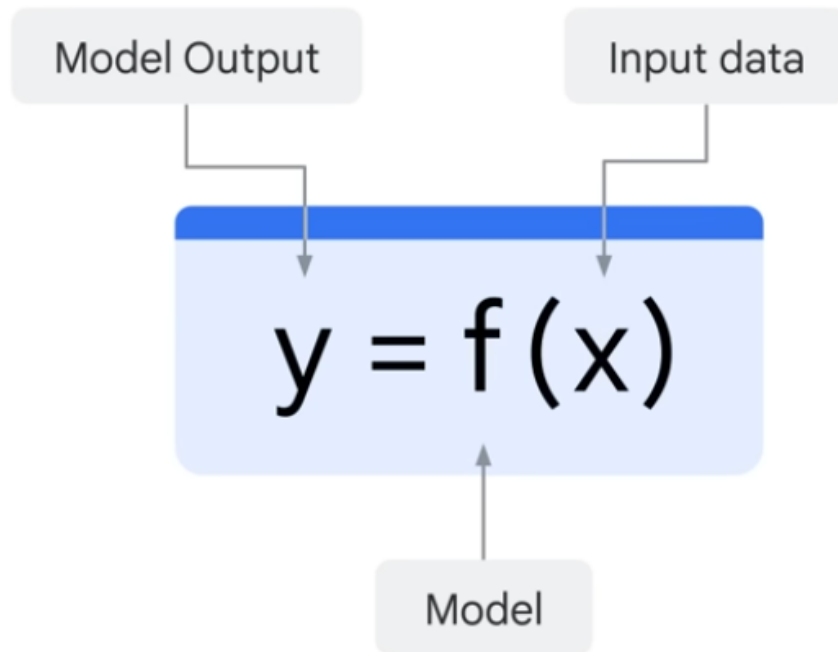
EC9640

Assessment/ Evaluation Details

Assessment Type	Assessment Method	Percentage
In-Course Assessment	Assignment	25
	Quiz	10
	Lab Report / Field Report	05
	Mid Semester Assessment	20
End of Course Evaluation	End Semester Examination	40

- Traditional AI
- Generative A

- Traditional AI systems are designed to respond to a particular set of inputs. But not only do they respond, but they can also learn from the data and make decisions and predictions based on that data. The most well-known example is if you're playing chess with a computer.
- Generative AI is an AI that can create new content, such as images, video and text, based on the data on which it is trained. It uses a sophisticated algorithm to create this new content based on prompts.



Not GenAI when y is a:

- Number
- Discrete
- Class
- Probability

Is GenAI when y is:

- Natural language
- Image
- Audio

Google Cloud Defines..

AI is the theory and development of computer systems able to **perform tasks normally requiring human intelligence.**

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Intended Learning Outcomes				
By the end of this course unit, the student should be able to				
- explain the fundamental principles of Artificial Intelligence; - Apply the basic principles, models, and algorithms of Artificial Intelligence to solve problems; - Devise an Artificial Intelligence based solution.				
Syllabus Outline				
Content	Hours			
	L	T	L/F	A
1. Introduction Fundamentals of Artificial Intelligence.	01			
2. Solving problems by searching Heuristic Search: A*; Optimization: Generate and Test, Simple Hill- Climbing, Steepest-Ascent Hill-Climbing; Adversarial Search: Games, Optimal Decisions in Games, Alpha-Beta Pruning.	06	02	03	
- Knowledge based system representation and inference • Propositional and Predicate logic; Inference in First-Order logic: Forward chaining, backward chaining, Constraint logic programming, Resolution; Knowledge Representation; Classical Planning;	06	02	03	
4. Expert Systems Introduction to Expert Systems; Architecture of Expert Systems; Applications of Expert Systems.	02			
- Natural Language Processing • Language Models; Applications: Text classification, Information Retrieval, Information Extraction.	03			
6. Artificial Intelligence Applications	02			18
	20	04	06	18

Thinking Humanly “The exciting new effort to make computers think . . . <i>machines with minds</i>, in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents.” (Poole <i>et al.</i>, 1998) “AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)



The cognitive modeling approach



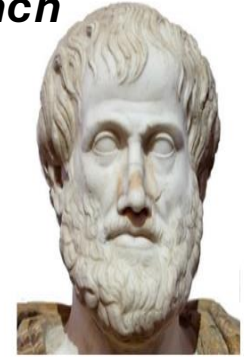
The Turing Test approach

The “laws of thought” approach



Essential Definition

Syllogism: a logical argument that applies deductive reasoning to arrive at a conclusion based on two or more propositions that are asserted, or assumed, to be true.



Aristotle

Classic Syllogism Structure:

P1: All the middle term (M) are the predicate (P).

P2: The subject (S) is the middle term (M).

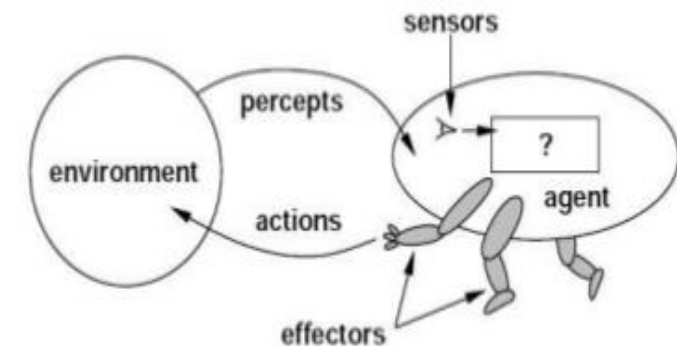
P3: Therefore, the subject (S) is the predicate (P).

Classic Syllogism Example:

P1: All men (M) are mortal (P).

P2: Socrates (S) is a man (M).

P3: Therefore, Socrates (S) is mortal (P).



The rational agent approach